

Flanker Task EEG functional data analysis

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Introduction

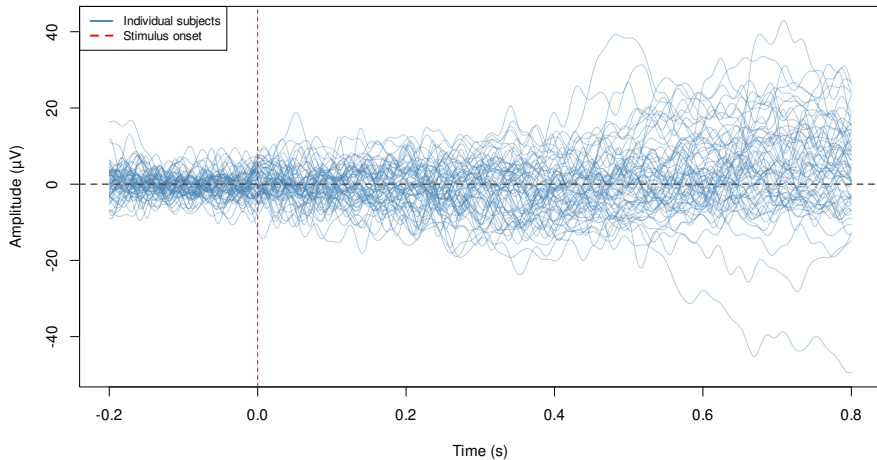
Project

- This dataset contains:
 - EEG brain recordings from 127 young adults.
 - Retrospective objective and subjective reports of childhood family socioeconomic status (SES).
 - SES indicators in adulthood.
 - Self-reports of ADHD symptoms.
- This project applies Functional Data Analysis methods to EEG data from 39 adults recorded during a Flanker task at electrode channel FC1.
- Data source - Cognitive Electrophysiology in Socioeconomic Context dataset (doi:10.18112/openneuro.ds006018.v1.2.2).

Data

Data

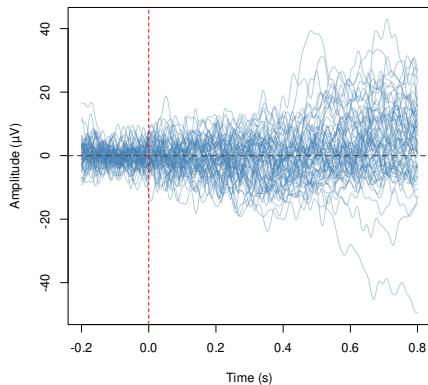
Flanker Task — Channel FC1 — All 62 Subjects



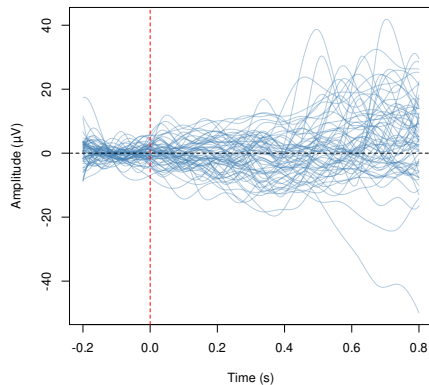
Smoothing

Smoothed data

Before Smoothing (raw)

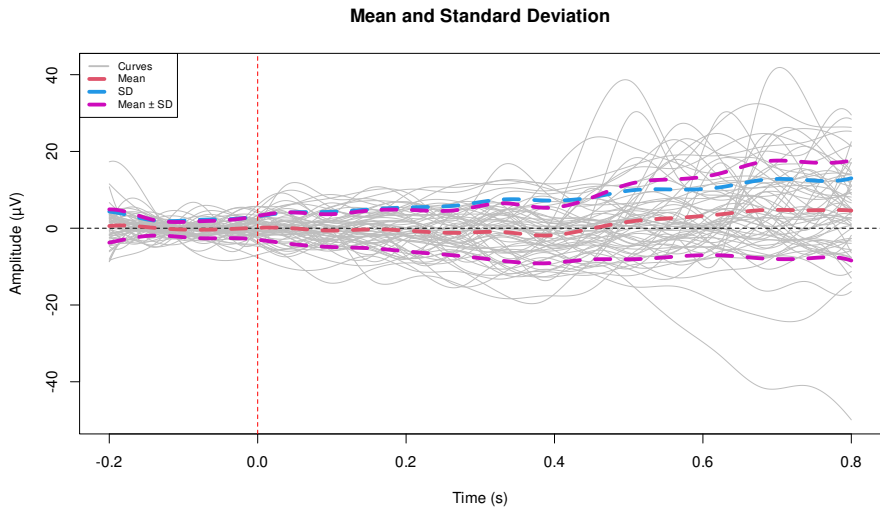


After Smoothing ($\lambda = 10^{-5}$)

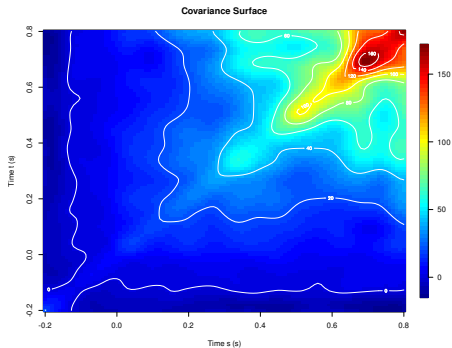
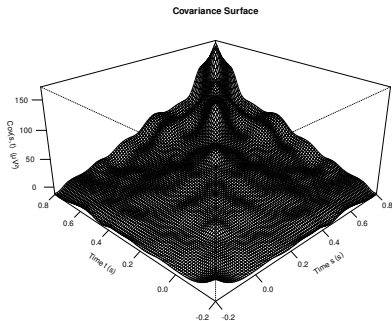


Exploratory Data Analysis

Mean and standard deviation

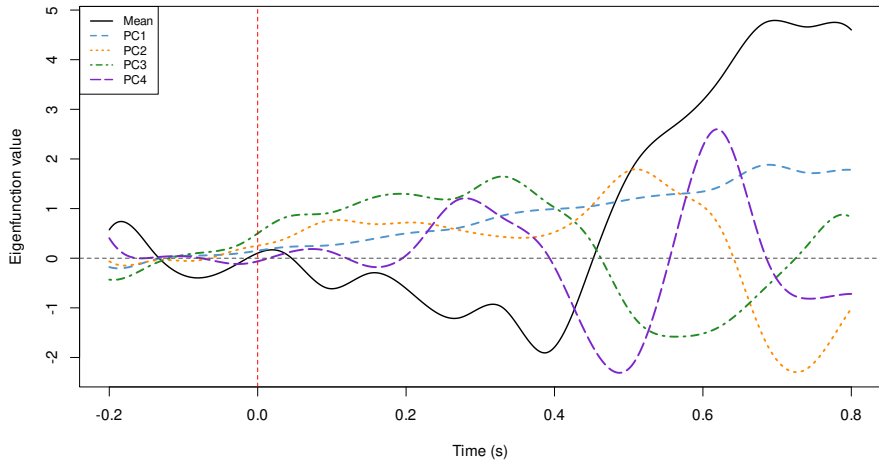


Covariance



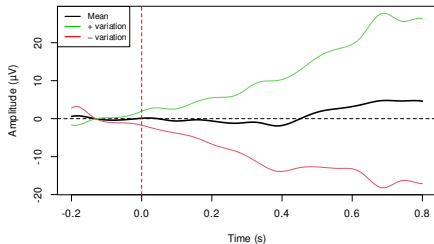
Principal Component Analysis I

PCA Harmonics and Mean Function

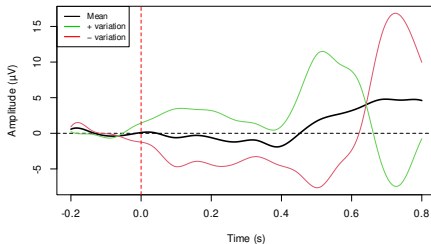


Principal Component Analysis II

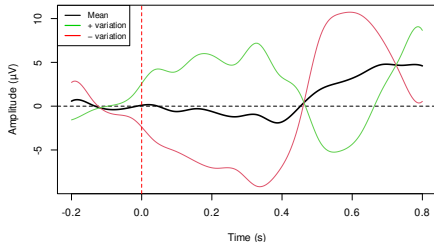
PC1 Perturbation (Expl. variance: 62.6%)



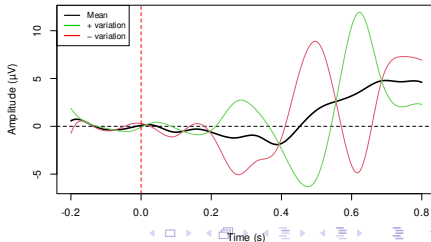
PC2 Perturbation (Expl. variance: 11.8%)



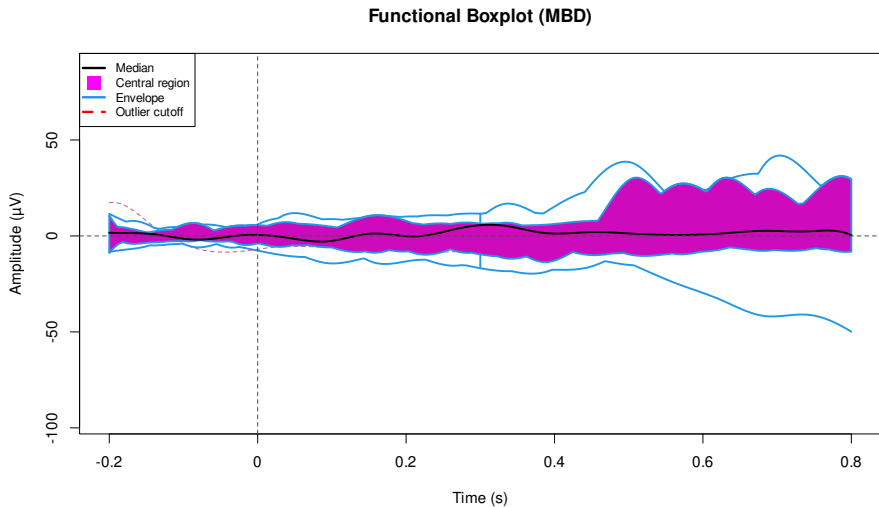
PC3 Perturbation (Expl. variance: 10.4%)



PC4 Perturbation (Expl. variance: 4.4%)

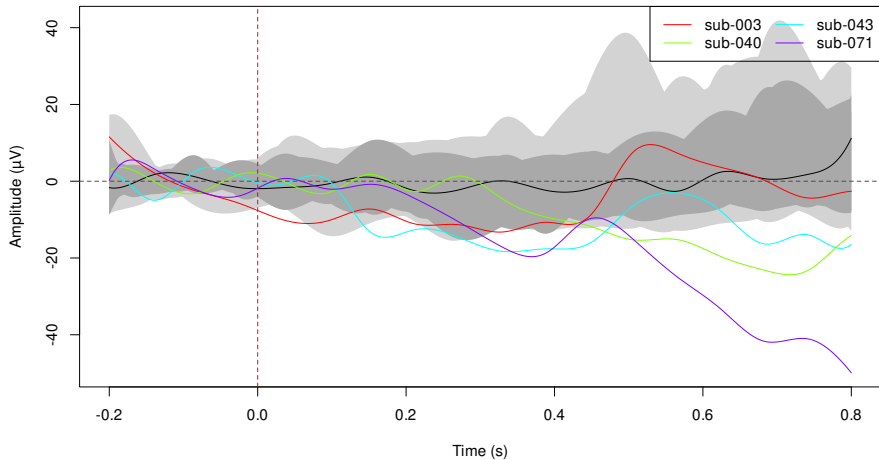


Functional Box Plot



Outliers

HDR Functional Boxplot, $\alpha = (0.05, 0.50)$



Thank you for your time!